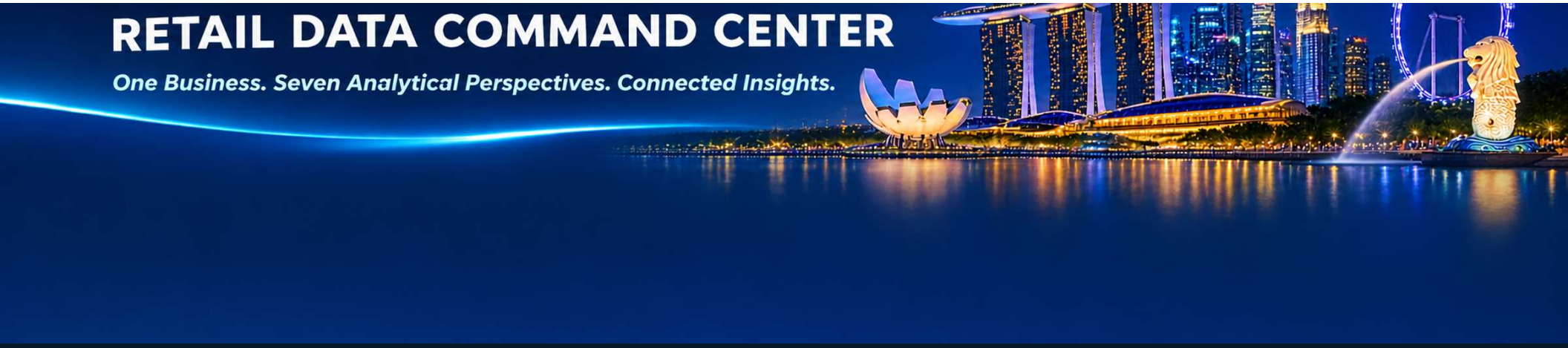


AN EXECUTIVE STORY

One business. Seven perspectives. One command center.

How MegaMart Singapore turns retail data into decisions — and where AI takes it next.

Durga Shankari

PORTFOLIO CASE STUDY • SYNTHETIC DATA • NOVEMBER 2026

Retail data is everywhere. Decisions should not be.

Fragmented visibility

Sales, customers, stock, delivery, campaigns and staffing are often analyzed separately.

Slow answers

Teams spend time locating, reconciling and explaining numbers instead of acting.

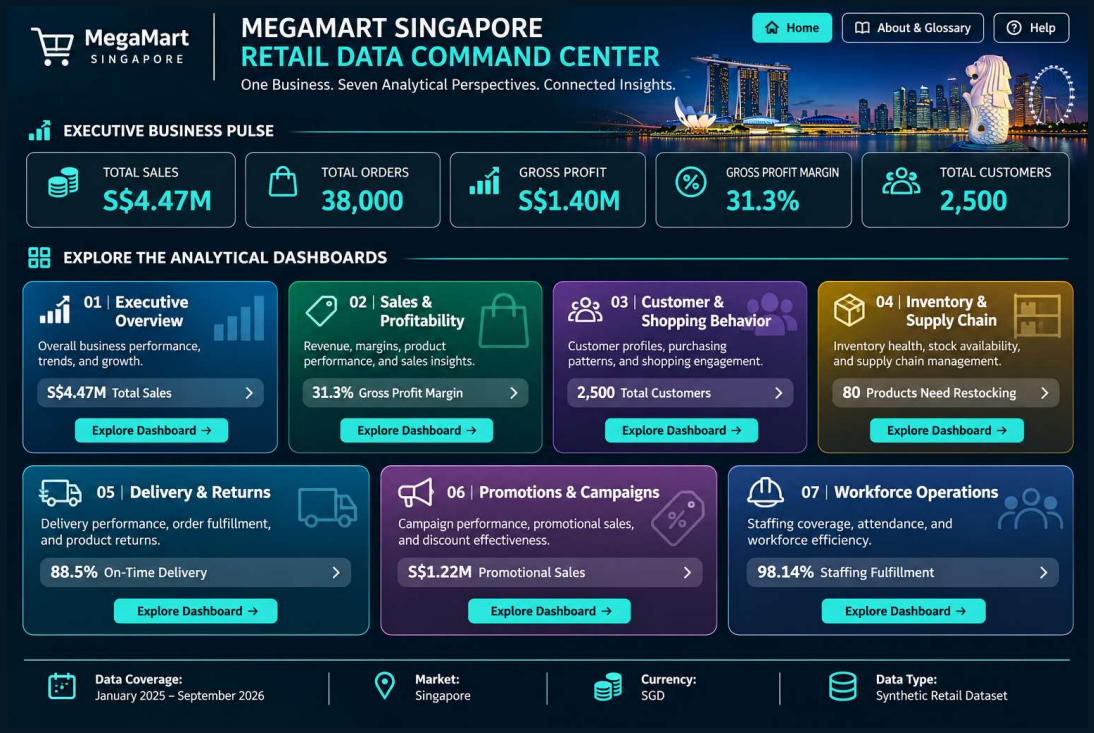
Disconnected action

An issue in one function can be invisible to the teams best placed to solve it.

THE OPPORTUNITY → A single entry point with connected, explainable analytical views.

Meet the Retail Data Command Center

A navigable executive cockpit — not another isolated dashboard.



ONE ENTRY POINT

7 analytical dashboards

+ About & Glossary

Custom cards, home navigation, consistent KPIs and contextual business insights.

Seven lenses. One retail story.

01

Executive Overview

Performance at a glance

02

Sales & Profitability

Where value is created

03

Customer & Shopping

Who buys and how

04

Inventory & Supply Chain

What is available

05

Delivery & Returns

How service performs

06

Promotions & Campaigns

What drives response

07

Workforce Operations

Who delivers the service

Five signals to orient every conversation

S$4.47M	38,000	S$1.40M	31.3%	2,500
TOTAL SALES	ORDERS	GROSS PROFIT	GROSS MARGIN	CUSTOMERS

The headline numbers provide context; the seven dashboards explain the drivers behind them.

From source records to trusted metrics

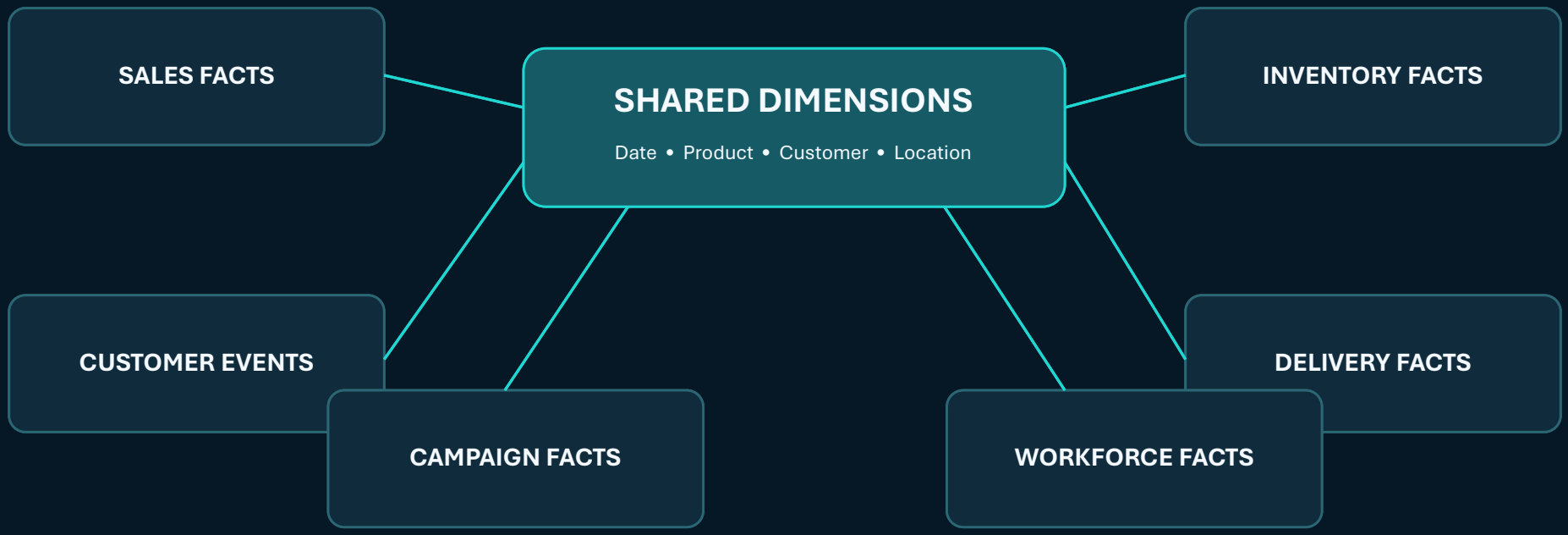
Conceptual data flow for the MegaMart demonstration — not a claim of a production data platform.



Enterprise evolution: automate ingestion, testing, lineage, refresh monitoring and access controls.

A shared model connects the story

Illustrative logical model: common keys and dimensions connect measures across domains.



A repeatable build–verify–refine cycle

01	**DESIGN**	Define questions, audience and KPIs.
02	**BUILD**	Create worksheets, filters and navigation.
03	**VERIFY**	Reconcile totals, formulas and breakdowns.
04	**EXPLAIN**	Add insights, glossary and interpretation notes.
05	**CONNECT**	Create the command center and test every route.

A beautiful dashboard is not enough. It must reconcile.

MEASURE

Validate the KPI formula, aggregation, units and formatting.

RECONCILE

Check totals across departments, shifts, days and filtered views.

EXPLAIN

Document rounding, definitions, filter scope and static insights.

Example: workforce fulfillment = 437,398.49 ÷ 445,689 = 98.14%.

Three signals worth investigating

31.3% gross margin

Profitability should be monitored alongside sales — revenue alone does not tell the whole story.

PROTECT VALUE

80 products flagged

Inventory signals can prioritize replenishment review and help avoid stock-related service issues.

PROTECT AVAILABILITY

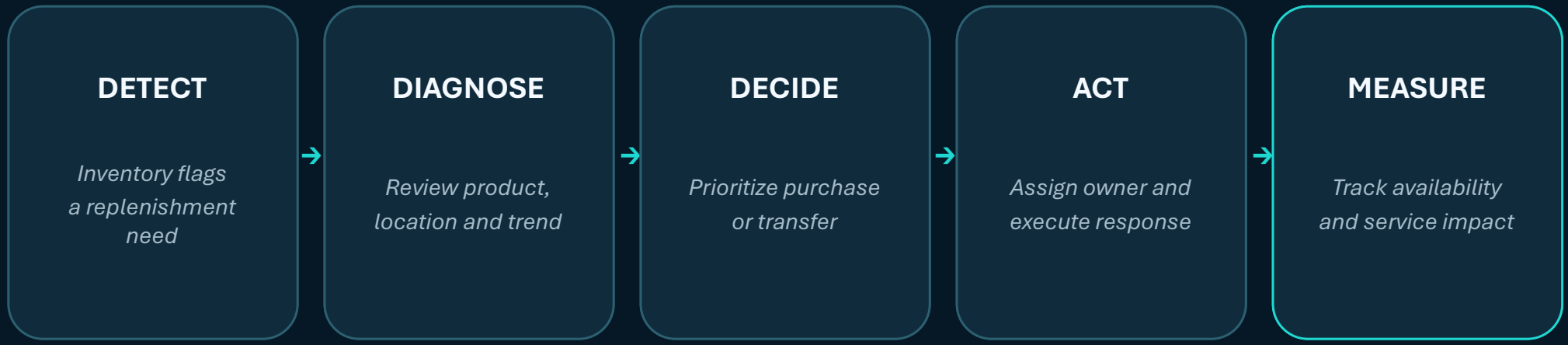
98.14% fulfillment

Staffing hours are close to plan, yet the 8,291-hour net gap merits investigation.

PROTECT CAPACITY

These are analytical prompts — not proof of root cause or financial impact.

Turn a signal into a decision



The command center helps people decide; operational systems and owners execute the action.

From monitoring the past to anticipating the next move

Proposed future capabilities — not features already implemented in the MegaMart prototype.

DEMAND FORECASTING

Predict product demand and anticipate stock needs.

CUSTOMER INTELLIGENCE

Recommend relevant products and identify churn risk.

OPERATIONS AI

Predict late deliveries and staffing pressure.

ANOMALY DETECTION

Flag unusual margin, return or transaction patterns.

AI should improve the customer journey — not just the forecast

01

DISCOVER

Relevant recommendations

02

BUY

Better stock visibility

03

RECEIVE

More reliable ETA

04

RETURN

Faster issue triage

05

STAY

Timely, useful engagement

Human oversight + consent and privacy controls + measurable customer outcomes

What an enterprise version must add

DATA GOVERNANCE

Owners, definitions, catalog, lineage

QUALITY & PIPELINES

Automated checks, refresh SLAs, monitoring

SECURITY & PRIVACY

Role-based access, encryption, minimization

MODEL GOVERNANCE

Validation, drift monitoring, human review

OPERATING MODEL

Incident response, adoption, change control

REGULATORY REVIEW

Applicable PDPA / GDPR obligations

A practical path from dashboard to intelligence

NOW

Trusted dashboards

Maintain QA, glossary and navigation.

NEXT

Connected operations

Integrate refreshed data, alerts and ownership.

THEN

Predictive pilots

Forecast demand; validate impact and bias.

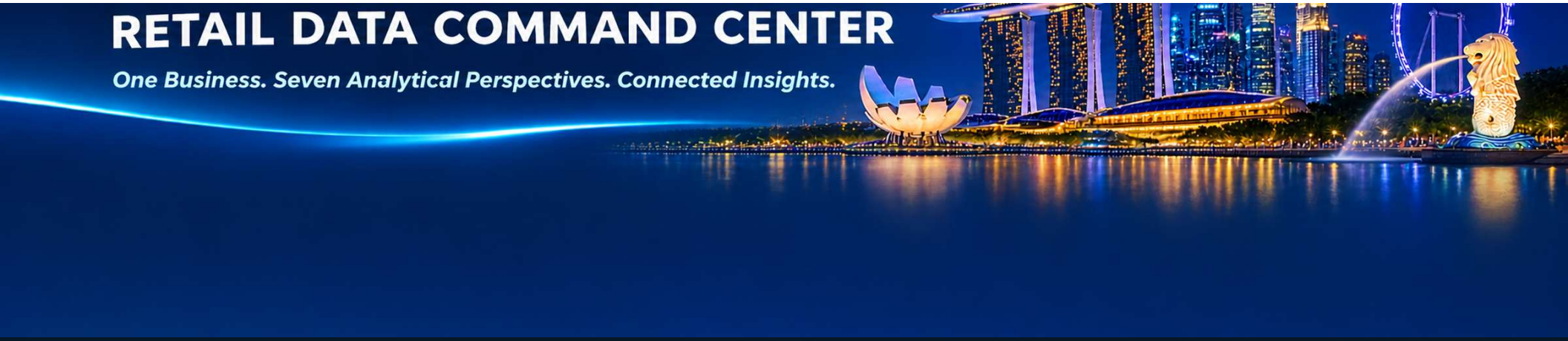
SCALE

Enterprise intelligence

Govern models, access and ongoing operations.

RETAIL DATA COMMAND CENTER

One Business. Seven Analytical Perspectives. Connected Insights.



THE BIG IDEA

A command center is not a screen.
It is a better way to run the business.

See clearly • Act faster • Learn continuously • Serve customers better

MegaMart Singapore | Synthetic-data portfolio demonstration